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1. SandboxAQ

Abstract

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Accurate and efficient prediction of protein–ligand binding free energies remains a barrier to the routine use of physics-based modeling in drug discovery. Although alchemical free energy methods can achieve high accuracy, their applicability is often limited by computational cost and chemical diversity. Here we present Nonequilibrium Chimeric Switching (NEX), a general free energy framework that leverages nonphysical states as endpoints in nonequilibrium transformations. In this work, we investigate one realization of NEX that integrates nonequilibrium statistical mechanics with a dual-topology alchemical transfer formulation for relative binding free energy calculations. By routing bidirectional nonequilibrium switches through a chimeric intermediate, NEX improves phase-space overlap and stabilizes free energy estimation. NEX achieves accurate and reproducible predictions across protein–ligand benchmarks, with most results within 1 kcal mol^{−1} of experiment. Comparative analysis shows that the chimeric intermediate is critical, reducing work-distribution variance, improving numerical stability, and outperforming direct nonequilibrium switching at equivalent

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Introduction

Binding free energy differences quantify molecular recognition and directly determine binding affinity, making them a critical objective in drug discovery. While candidate molecules can be evaluated entirely through synthesis and experimental screening, the combinatorial scale of chemical space has driven increased reliance on computational methods to prioritize compounds.^{1,2} Despite decades of progress, the accurate prediction of binding free energies *in silico* remains a bottleneck: existing approaches struggle to achieve the precision, efficiency, and generality required to support end-to-end drug discovery pipelines.^{3,4}

Physics-based alchemical methods, such as free energy perturbation (FEP), are the gold standard for computational free energy prediction,⁵⁻⁸ but their applicability is limited by the need for extensive sampling^{9,10} and favorable phase-space overlap between states.¹¹⁻¹³ These limitations become acute for flexible systems and chemically diverse transformations, where sampling costs increase and prediction accuracies can degrade.^{3,4,14} Machine learning methods enable rapid inference, but their predictive reliability is constrained by the scope of their training data, limiting their ability to generalize to novel chemical space.^{15,16} As a result, there is a clear motivation for first-principles methods that maintain physical rigor while achieving efficient scalability across diverse high-throughput settings.

A range of alchemical methods have been developed to address specific use cases that can be broadly classified as yielding absolute or relative free energy predictions.¹⁷⁻²⁰ Absolute methods estimate the free energy of an individual molecule, while relative methods compute the relative difference in free energy between two molecules.^{7,8} Methods that use absolute free

energy estimation can be expensive, but enable larger chemical changes and hit finding,^{21,22} while relative FEP methods are utilized more frequently for lead optimization with minimal scaffold perturbation.^{3,4,23} More recent advances offer increased flexibility, supporting both absolute and relative calculations,^{24–26} allowing more accurate scaffold hopping,²⁷ or performing absolute transformations at rapid speed but with accuracy losses for hit finding.^{7,28} Despite these advances, many existing methods remain limited by computational cost and are not readily scalable to large compound libraries.^{29,30}

Nonequilibrium statistical mechanics has fundamentally reshaped approaches to free energy estimation, enabling access to equilibrium thermodynamic quantities without requiring equilibrium sampling. Central to this framework are fluctuation theorems, which provide exact relationships between nonequilibrium work and equilibrium free energy differences. The Jarzynski equality³¹ relates the exponential average of work performed along finite-time, nonequilibrium trajectories to the equilibrium free energy difference, while the Crooks fluctuation theorem³² establishes a direct connection between forward and reverse work distributions. In practice, the Crooks formalism is often preferred in alchemical free energy calculations due to its improved statistical efficiency under finite sampling.^{30,33,34}

Importantly, these theoretical advances have been directly supported by experimental validation. In single-molecule force spectroscopy, techniques such as atomic force microscopy and optical tweezers exploit fluctuation theorems to reconstruct equilibrium free energy landscapes from irreversible pulling experiments.^{35–37} These results demonstrate that nonequilibrium measurements can yield accurate equilibrium thermodynamic quantities, establishing nonequilibrium approaches as both a rigorous and experimentally validated framework for free energy estimation. However, experimental approaches are limited to what is physically feasible. In simulations, we are enabled to develop methods that break physical reality as long as statistical mechanics is obeyed.

Here, we introduce Nonequilibrium Chimeric Switching (NEX), a free energy method that rethinks how nonequilibrium alchemical transformations are performed in simulation. NEX

uses nonequilibrium statistical mechanics to estimate free-energy differences from ensembles of switching trajectories initiated from equilibrium configurations.^{31,32,38} Nonequilibrium methods reduce the amount of sampling required during an alchemical transformation.^{33,39,40} While nonequilibrium approaches have been applied to lead optimization with R-group modifications,³⁰ applicability is limited by significant structural overlap.⁴¹ Nonequilibrium methods were previously extended to frameworks in which whole molecules are transformed for absolute binding free energy calculations,³⁴ but encountered challenges with large alchemical distances, leading to slow convergence and long sampling times.

For RBFE calculations, NEX employs a dual topology framework,⁴² in which both ligand topologies are fully preserved throughout the simulation. This formulation enables the treatment of chemically diverse transformations, including scaffold hopping, charge changes, ring-breaking, and substantial conformational rearrangements, with cleanly separable ligand partition functions, which is a known challenge in single-topology approaches.⁴³

A key innovation of NEX is the use of chimeric intermediate states to launch nonequilibrium sampling. These chimeric states represent nonphysical hybrids, such as alchemical states, that interpolate between endpoint systems and reshape the alchemical pathway. We demonstrate that this framework improves phase-space overlap during nonequilibrium switching, reduces variance in work distributions, and stabilizes free energy estimates. Our results indicate that NEX is positioned to address a longstanding challenge in RBFE calculations: efficiently connecting chemically distant states without prohibitive sampling costs. To our knowledge, this is the first nonequilibrium switching method to systematically use both physical and nonphysical endpoints for free energy estimation.

We demonstrate NEX’s performance on a range of protein–ligand RBFE benchmarks,⁴⁴ showing that chimeric switching yields accurate and reproducible predictions across diverse targets. Through controlled comparisons, we show that chimeric intermediates aid both accuracy and numerical stability, significantly outperforming direct nonequilibrium switching approaches with equivalent sampling time. Mechanistic analysis reveals that these im-

provements arise from reduced work variance and improved overlap between forward and reverse processes.^{38,45} Finally, we demonstrate that NEX provides competitive performance on chemically challenging scaffold-hopping transformations, a regime where conventional RBFE methods often fail.^{3,27} Together, these results position NEX as a scalable and general framework for free energy prediction, with the potential to substantially expand the practical scope of RBFE calculations in drug discovery.

Methods

Benchmark Systems

To evaluate the performance and accuracy of the Nonequilibrium Chimeric Switching (NEX) method for lead optimization transformations, we utilized a well-established benchmark set of protein-ligand systems originally curated by Wang et al.^{44,46,47} This set, comprising eight diverse protein targets (MCL-1, TYK2, JNK1, PTP1B, BACE, Thrombin, CDK2, and p38), was chosen because it represents a standard for evaluating relative binding free energy protocols. Each ligand series spans a range of chemical moieties and binding affinities, providing a rigorous test of new alchemical methods. These systems are particularly suitable as they have been extensively studied in the literature, providing a rich dataset for comparative analysis and validation. From this benchmark set, we evaluated performance on TYK2, CDK2, JNK1, Thrombin, BACE1, and p38. To further stress-test the method in the context of hit-to-lead optimization, we benchmarked NEX against a previously studied BACE1 set that includes scaffold hopping transformations between ligand series, which can be challenging for standard RBFE methods.²⁷ The relevant perturbation maps, input files, and free energy values were collected from the Supporting Information of Baumann et al.²⁷

Alchemical Protocol

The methodology we employed here is one realization of NEX and was developed to study RBFE transformations, in which larger chemical changes may be possible. The approach studied here uses a dual topology alchemical protocol inspired by the Alchemical Transfer Method (ATM).^{24–26,48} In this approach, both the starting ligand (A) and the ending ligand (B) are present in the simulation box simultaneously. One ligand is located in the protein’s binding pocket, while the other is placed in the bulk solvent. The alchemical transformation then effectively swaps their positions. The nonbonded forces were implemented using an `openmm.ATMForce` object from `OpenMM` version 8.1.2.⁴⁹ The total potential $U(\lambda)$ is the sum of the reference potential and a generalized softplus alchemical perturbation function $W_\lambda(u)$:

$$U(\lambda) = U_0 + W_\lambda(u) \quad (1)$$

where U_0 and U_1 are the potential energies of the system at the non-alchemical endpoint states, respectively, $u = U_1 - U_0$ is the perturbation energy, and $W_\lambda(u)$ is defined as:

$$W_\lambda(u) = \frac{\lambda_2 - \lambda_1}{\alpha} \ln (1 + e^{-\alpha(u_{sc}(u) - U_h)}) + \lambda_2 u_{sc}(u) + W_0 \quad (2)$$

The soft-core potential, $u_{sc}(u)$, smoothly caps the perturbation energy to prevent singularities:

$$u_{sc}(u) = \begin{cases} u & \text{if } u < U_{b,core} \\ (U_{max} - U_{b,core})f_{sc} + U_{b,core} & \text{if } u \geq U_{b,core} \end{cases} \quad (3)$$

with the scaling function f_{sc} defined by:

$$f_{sc} = \frac{z^{A_{core}} - 1}{z^{A_{core}} + 1} \quad (4)$$

$$z = 1 + \frac{2y}{A_{core}} + 2 \left(\frac{y}{A_{core}} \right)^2 \quad (5)$$

$$y = \frac{u - U_{b,core}}{U_{max} - U_{b,core}} \quad (6)$$

Default parameter values were set to $A_{core} = 0.0625$, $\alpha = 0.0239 \text{ (kJ/mol)}^{-1}$, $U_{b,core} = 418.4 \text{ kJ/mol}$, $U_h = 460.24 \text{ kJ/mol}$, $U_{max} = 836.8 \text{ kJ/mol}$, and $W_0 = 0 \text{ kJ/mol}$.²⁵

The parameters λ_1 and λ_2 are functions of the alchemical progress variable λ , which spans from 0 (physical state) to 1 (chimeric state) for each leg of the calculation. A ‘direction’ parameter distinguishes the two legs: the first leg (ligand A bound/B unbound $\rightarrow$ intermediate) uses ‘direction = +1’, while the second leg (ligand B bound/A unbound $\rightarrow$ intermediate) uses ‘direction = -1’, which effectively redefines the perturbation energy as $u = U_0 - U_1$. For both legs, the schedule for λ_1 and λ_2 as a function of $\lambda \in [0, 1]$ is given by the following piecewise protocol:

$$\lambda_1(\lambda) = \begin{cases} 0 & 0 \leq \lambda \leq 0.5 \\ \lambda - 0.5 & 0.5 < \lambda \leq 1 \end{cases} \quad (7)$$

$$\lambda_2(\lambda) = \begin{cases} \lambda & 0 \leq \lambda \leq 0.5 \\ 0.5 & 0.5 < \lambda \leq 1 \end{cases} \quad (8)$$

We chose this ATM-like framework for several key reasons:

- **Scaffold Hopping:** By avoiding a direct atom-to-atom mapping between ligands, this protocol naturally accommodates significant structural differences, such as ring-breaking/forming events or changes in the core scaffold.^{46,50,51}
- **Charge-Changing Transformations:** The entire system maintains a constant net

charge throughout the simulation because both ligands are always present in the simulation box. This circumvents the need for complex corrections that arise in single-topology methods when transforming between ligands with different net charges.^{46,50,51}

- **Simplified Setup:** An ATM-like approach eliminates the need for complex topologies with dummy atoms or soft-core potentials, streamlining the system preparation process²⁴ and ensuring partition function separability.⁴³
- **Force field agnostic:** Since alchemical transfer performs energy interpolation rather than parameter interpolation, arbitrary energy functions can be readily deployed.^{24,52,53}

Molecular Dynamics Simulations

System Preparation

Force field parameters were assigned as follows: Amber’s **ff14SB** for the protein,⁵⁴ OpenForceField’s **openff-1.3.1** (Parsley) for the small molecules⁵⁵ with charges derived from the **AM1-BCC** model,⁵⁶ and the **TIP3P** model for water molecules.⁵⁷ To enable a longer integration timestep, water molecules were treated as rigid, and all bonds between heavy atoms and hydrogen atoms for both the protein and the ligands were constrained.

To prevent conformational drift, a flat-bottom-to-quadratic potential ($25 \text{ kcal} \cdot \text{mol}^{-1} \cdot \text{\AA}^{-2}$) with a radius of $1.5 \text{ \AA}$ was applied to all receptor α -carbons. A similar potential was used to restrain the center of mass of both ligands to the binding site, defined by the α -carbons of the binding site residues, using a force constant of $25 \text{ kcal} \cdot \text{mol}^{-1} \cdot \text{\AA}^{-2}$ and a radius of $5 \text{ \AA}$. To maintain co-orientation between the two ligands, Boresch-like⁵⁸ alignment restraints were applied to triples of ligand heavy atoms using the same formalism as applied in ATM.^{24,25,46} These included a distance restraint with a spring constant of $2.5 \text{ kcal} \cdot \text{mol}^{-1} \cdot \text{\AA}^{-2}$, and angle and dihedral restraints, each with a stiffness of $25 \text{ kcal mol}^{-1} \text{ radian}^{-2}$.

All simulations were conducted under ambient conditions (i.e., the temperature maintained at 298.15 K and pressure at 1 atm). The system utilized a Langevin integrator^{59,60} (i.e.

`openmm.LangevinMiddleIntegrator`) with a collision frequency of 1 ps^{-1} and a simulation time step of 2 fs for production simulations.

The system was prepared by placing the starting ligand (A) in the receptor’s binding pocket, while the other ligand (B) was placed in the bulk solvent at a displacement of 38.0 Å. The simulation box was populated with TIP3P water molecules and a 0.15 M KCl buffer to model physiological conditions.

Minimization and Equilibration

The system underwent a multi-stage equilibration protocol to ensure it reached thermal and structural equilibrium prior to production runs. Flat-bottom positional restraints with a force constant of $25.0 \text{ kcal} \cdot \text{mol}^{-1} \cdot \text{Å}^{-2}$ and a radius of 1.5 Å were applied to all heavy atoms during energy minimization, thermal annealing, thermalization, λ -annealing, and NPT equilibration in the alchemical intermediate ($\lambda = 0.5$) state. The initial system configuration was first energy minimized using OpenMM’s `simulation.minimizeEnergy()` method with a convergence tolerance of $10.0 \text{ kcal} \cdot \text{mol}^{-1} \cdot \text{Å}^{-1}$, which performs a local minimization using the L-BFGS algorithm.^{61,62}

Following this, the system’s temperature was linearly annealed from 50.0 K to a final target temperature of 298.15 K over 25,000 steps, corresponding to a total simulation time of 50 ps. This temperature was incremented every 10 ps.

Succeeding the temperature annealing, the system was equilibrated at the constant pressure and temperature ensemble (NPT) at a target temperature of 298.15K and a pressure of 1 atm for 150 ps. Pressure control was maintained using a Monte Carlo Barostat with an update frequency of 25 steps.

The system was then linearly annealed to the alchemical intermediate $\lambda = 0.5$ over 250ps, with λ being incremented every 10 ps. The temperature and pressure were maintained. The soft core parameters were set to $U_{max} = 1000 \text{ kcal} \cdot \text{mol}^{-1}$, $U_0 = 500 \text{ kcal} \cdot \text{mol}^{-1}$, and the scaling parameter $\alpha = 0.0625$ during this stage. Finally, the system was subjected to a final

equilibration run of 150 ps in this alchemical intermediate.

The relaxed coordinates collected at the end of this multi-stage equilibration protocol were used as initial coordinates for subsequent production simulations. The temporary, positional restraints placed on the system during the preceding stages were also removed prior to subsequent production runs.

NEX Protocol

The core of the NEX production protocol developed here can be divided into three distinct phases. First, the system was equilibrated at the alchemical end-points ($\lambda = 0$ and $\lambda = 1$) and the intermediate state ($\lambda = 0.5$). These initial equilibration (warm-up) simulations were run for 1.25 ns each. Following the warm-up, three independent production trajectories were launched from the final configurations of each of the three equilibrated λ states ($\lambda = 0$, $\lambda = 0.5$, and $\lambda = 1$). Each trajectory was run for 25 ns, during which 100 snapshots were collected at an interval of 0.25 ns. This yielded a total of 300 statistically independent snapshots per alchemical state to serve as starting configurations for subsequent nonequilibrium work calculations. Finally, we execute fast non-equilibrium alchemical switches using the `openmmtools`’s `AlchemicalNonequilibriumLangevinIntegrator`⁶³ with a `V R H O R` `V` splitting scheme.⁶⁴ `V`, `R` represent deterministic velocity and position updates respectively. While, `O` represents an Ornstein-Uhlenbeck⁶⁵ step, i.e a stochastic update of velocities to simulate interactions with a heat bath. `H` represents the Hamiltonian update, which linearly anneals λ over the duration of the trajectory. The alchemical coupling parameter, λ , was linearly annealed over a fixed duration of 100 ps per switch.

Non-equilibrium switches were then executed between the alchemical end-points ($\lambda = 0$, $\lambda = 1$) and the intermediate state ($\lambda = 0.5$), initiated from the 300 respective snapshots collected for each state. This protocol was divided into two main legs:

1. Switches were performed between the $\lambda = 0$ and $\lambda = 0.5$ states, comprising $\lambda = 0 \rightarrow \lambda = 0.5$ and $\lambda = 0.5 \rightarrow \lambda = 0$ (forward/reverse) transitions.

2. Switches were performed between the $\lambda = 1$ and $\lambda = 0.5$ states, comprising $\lambda = 1 \rightarrow \lambda = 0.5$ and $\lambda = 0.5 \rightarrow \lambda = 1$ (forward/reverse) transitions.

The process of switching between states in an RBFE process is conceptually illustrated in **Figure 1A**, which shows a transformation between an alchemical chimeric state and the two physical end states. To interrogate the benefit of the chimeric state, direct switches were also performed between the alchemical end-points ($\lambda = 0 \rightleftharpoons \lambda = 1$), eschewing the chimeric alchemical intermediate ($\lambda = 0.5$).

Analysis

The Bennett Acceptance Ratio (BAR) method^{38,66} was applied to the forward and reverse nonequilibrium work distributions for each leg to determine the individual free energy differences (ΔG) and their asymptotic uncertainties (σ). The collected work values from the $\lambda = 0 \rightarrow 0.5$ and $\lambda = 0.5 \rightarrow 0$ switches were used to calculate $\Delta G_{0 \rightarrow 0.5}$. Similarly, collected work values from the $\lambda = 1 \rightarrow 0.5$ and $\lambda = 0.5 \rightarrow 1$ switches were used to calculate $\Delta G_{1 \rightarrow 0.5}$. All numerical solutions for the BAR equations and subsequent uncertainty estimations were executed using the `pymbar` Python library.^{67,68}

The final relative free energy, $\Delta\Delta G$, representing the complete alchemical transformation from Ligand 1 ($\lambda = 0$) to Ligand 2 ($\lambda = 1$), was calculated by summing the free energy differences of the two legs: $\Delta\Delta G = \Delta G_{0 \rightarrow 0.5} + \Delta G_{0.5 \rightarrow 1}$. The total uncertainty ($\delta(\Delta\Delta G)$) was obtained by summing the variances (σ^2) of the individual leg estimates in quadrature, $\delta(\Delta\Delta G) = \sqrt{\sigma_{Leg1}^2 + \sigma_{Leg2}^2}$. In cases where an alchemical intermediate was not included, a single BAR calculation was performed over the entire alchemical path ($\lambda = 0 \rightleftharpoons \lambda = 1$).

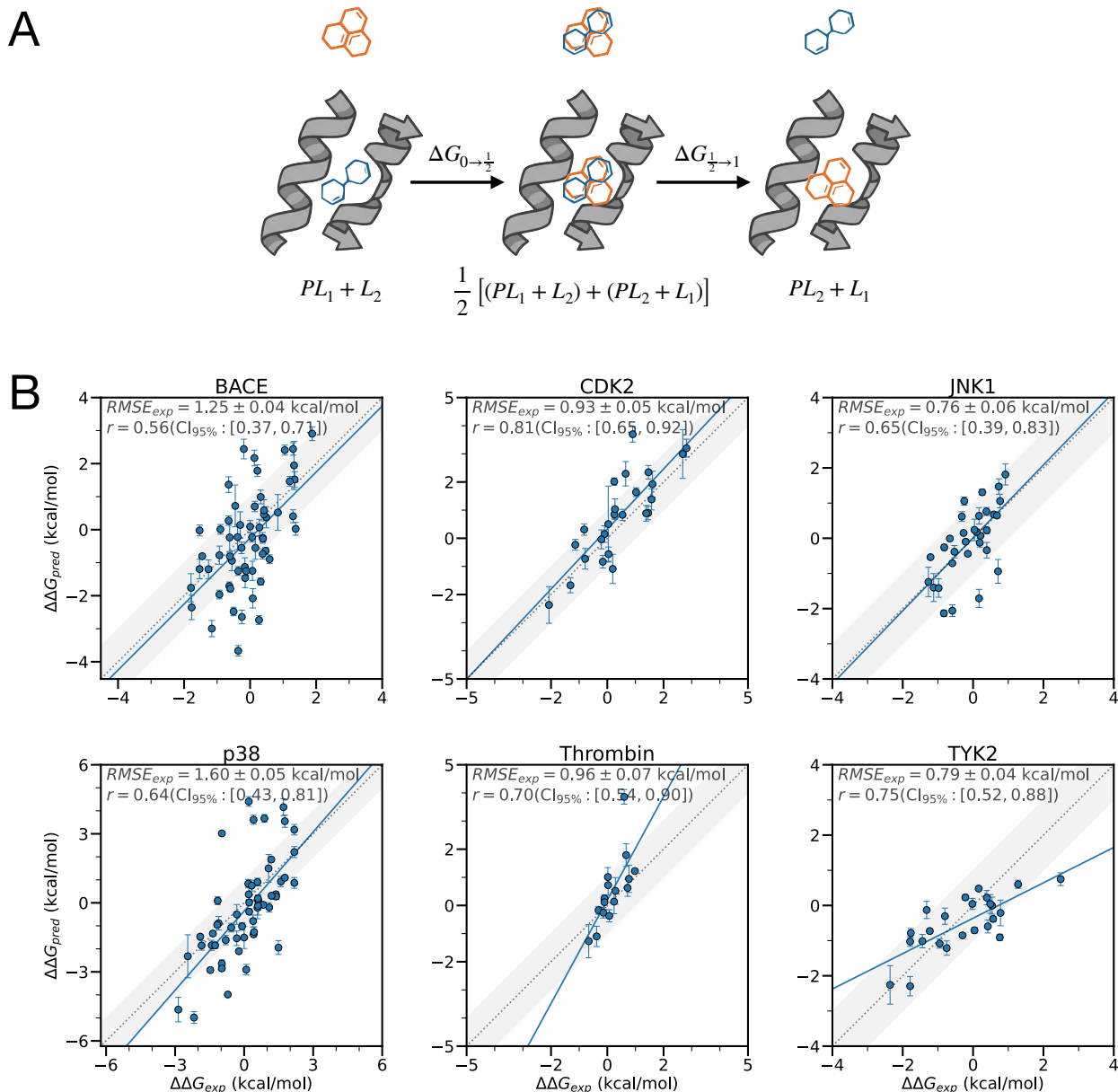


Figure 1: Nonequilibrium Chimeric Switching lead optimization benchmarks correlate well with experimental data. (A) Schematic of an alchemical chimeric switch between three states: the initial physical state, the intermediate chimeric state, and the final physical state. (B) Predicted ($\Delta\Delta G_{pred}$) versus experimental ($\Delta\Delta G_{exp}$) relative binding free energies for six protein targets (in kcal/mol). Blue circles represent the average prediction for an individual calculation, with error bars indicating the standard error of the mean (SEM) from three independent runs. The dotted line indicates perfect agreement, the solid blue line is the linear regression fit, and the shaded gray region denotes a ± 1.0 kcal/mol deviation. Insets provide the Root Mean Square Error ($RMSE_{exp}$) and Pearson correlation coefficient (r) with 95% confidence intervals.

Results

Chimeric Intermediates Yield Accurate Binding Affinity Predictions For Lead Optimization

We evaluated the predictive accuracy of the NEX RBFE protocol on a diverse lead optimization benchmark comprising six well-studied protein targets: BACE, CDK2, JNK1, p38, Thrombin, and TYK2. As illustrated in Fig. 1A, the method employs a three-state nonequilibrium switching protocol, in which the transformation between physical endpoint states is decomposed into serial and bidirectional nonequilibrium switches via a chimeric intermediate. The first set of bidirectional switches, $\Delta G_{0 \rightarrow 0.5}$ and *vice versa*, drives the system between a physical state, where the initial ligand is bound and the final ligand resides in solvent, into a hybrid state in which both ligands interact simultaneously with the protein and solvent. The second set of switches, $\Delta G_{0.5 \rightarrow 1}$, completes the thermodynamic cycle to the final physical state.

Across all systems (Fig. 1B), NEX yields predicted relative binding free energies that are in good agreement with experiment, with most predictions falling within the ± 1 kcal/mol error envelope. Quantitatively, the root mean square error (RMSE_{exp}) relative to experiment ranged from 0.76 kcal/mol (JNK1) to 1.60 kcal/mol (p38), while Pearson correlation coefficients (r) spanned from 0.56 (BACE) to 0.81 (CDK2). While performance was generally strong, some system-specific variability was observed; targets such as CDK2 and Thrombin appeared particularly well-modeled, whereas BACE and p38 showed slightly greater scatter in their predictions compared to experimental data. Importantly, for all systems, the 95% confidence intervals confirmed a statistically significant linear relationship between prediction and experiment.

Predictions exhibit consistently low statistical uncertainty, as reflected by small error bars (standard error of the mean across three independent runs), indicating high precision and reproducibility in the computed $\Delta\Delta G$ values. Compared to reported equilibrium

methods,⁴⁶ the NEX protocol achieves comparable or improved precision despite relying on nonequilibrium sampling. In addition, the chimeric switching protocol demonstrates strong numerical stability across all systems, with negligible failure rates (NaNs observed in only 0–2% of runs), supporting robust convergence even for challenging transformations. The benchmark set consists of congeneric ligand series with no net charge changes between ligand pairs, providing a controlled environment to assess the efficacy of the chimeric switching protocol. Across these varied protein binding sites and ligand sets, these results indicate that NEX can deliver both high precision and simulation stability, despite the previously reported challenges associated with nonequilibrium approaches where entire molecules are transformed.³⁴

Chimeric states improve the accuracy of binding free energy predictions in a dual topology nonequilibrium approach

To isolate and quantify the impact of the alchemical chimeric intermediate within the NEX protocol, we performed a comparative set of standard end-to-end nonequilibrium free energy calculations with results reported in **Figure 2**. The same NEX alchemical protocol from **Figure 1** with equivalent sampling time, but the chimeric intermediate state was omitted. Nonequilibrium switches were performed directly between the two physical end states (**Figure 2A**). In this nonchimeric approach, a single switch ($\Delta G_{0 \rightarrow 1}$) transitions directly from a state corresponding to a protein-bound starting ligand (with the ending ligand in bulk solvent) to the final exchanged state. To ensure a comparable computational budget, the nonequilibrium switching time was doubled to 200 ps. As a control, to isolate the performance comparison strictly to the inclusion of the chimeric state, each nonequilibrium switch was seeded from the pre-equilibrated conformations at the physical end states generated for the simulations in **Figure 1**.

To evaluate the impact of the nonchimeric modification, we performed a pairwise statistical comparison between the original chimeric protocol (M1) and the nonchimeric method

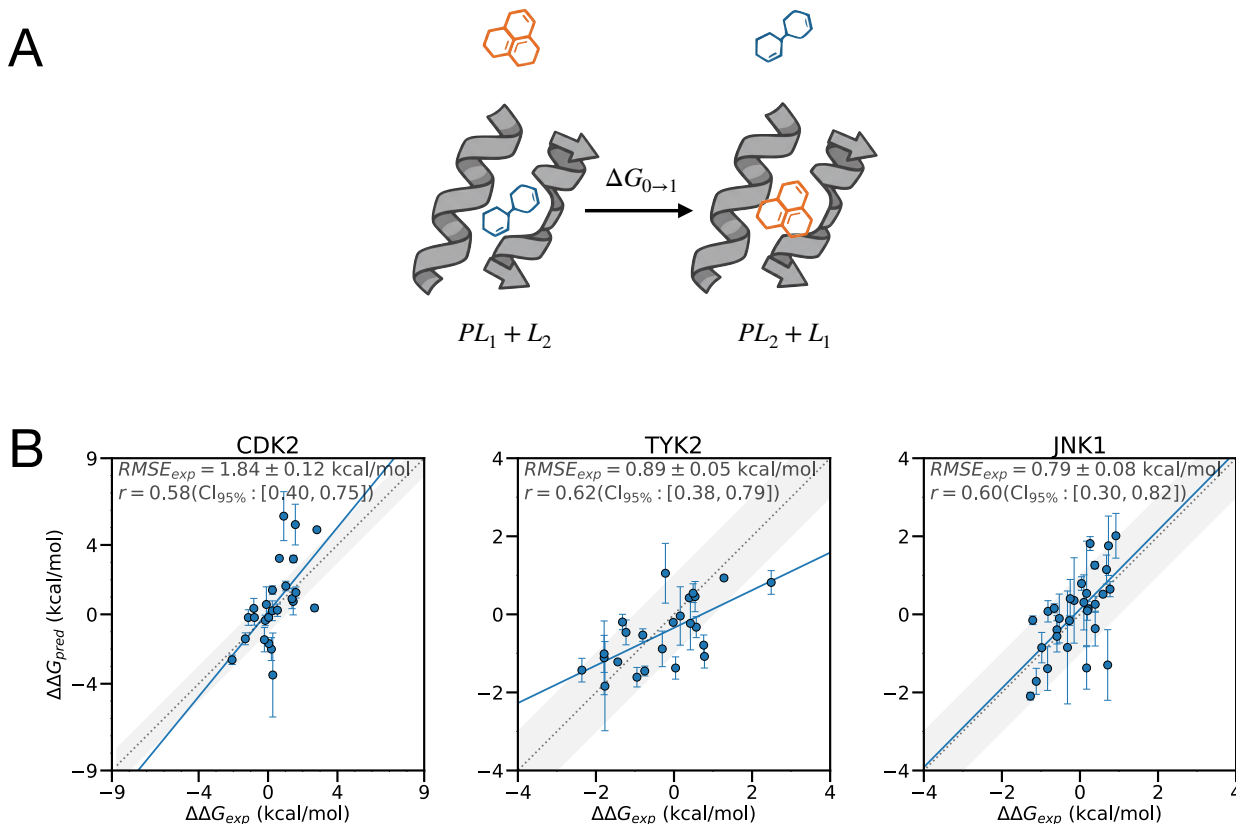


Figure 2: **Nonequilibrium switching without a chimeric intermediate shows degraded predictive performance.** (A) Schematic of an alchemical nonchimeric switch transitioning directly between two physical end states. (B) Predicted versus experimental relative binding free energies for a subset of the protein-ligand systems from **Figure 1**. Plotting conventions, including axes, error bars, regression lines, and insets, are identical to those described in **Figure 1**.

(M2) across three protein systems (Table 1). Analysis of the CDK2 system ($N = 25$) revealed a statistically significant degradation in performance when using the nonchimeric method. The original chimeric approach yielded a higher Pearson correlation ($\Delta r = 0.217$, 95% CI [0.068, 0.363]) and a lower root-mean-square error ($\Delta RMSE = -0.863$, 95% CI [-1.287, -0.422]), a difference confirmed by a Wilcoxon signed-rank test ($p = 0.0074$). Conversely, for the TYK2 ($N = 24$) and JNK1 ($N = 30$) systems, the nonchimeric method did not result in a statistically distinguishable change in accuracy. In both cases, the 95% confidence intervals for Δr and $\Delta RMSE$ spanned zero, and p -values exceeded the significance threshold ($p > 0.05$).

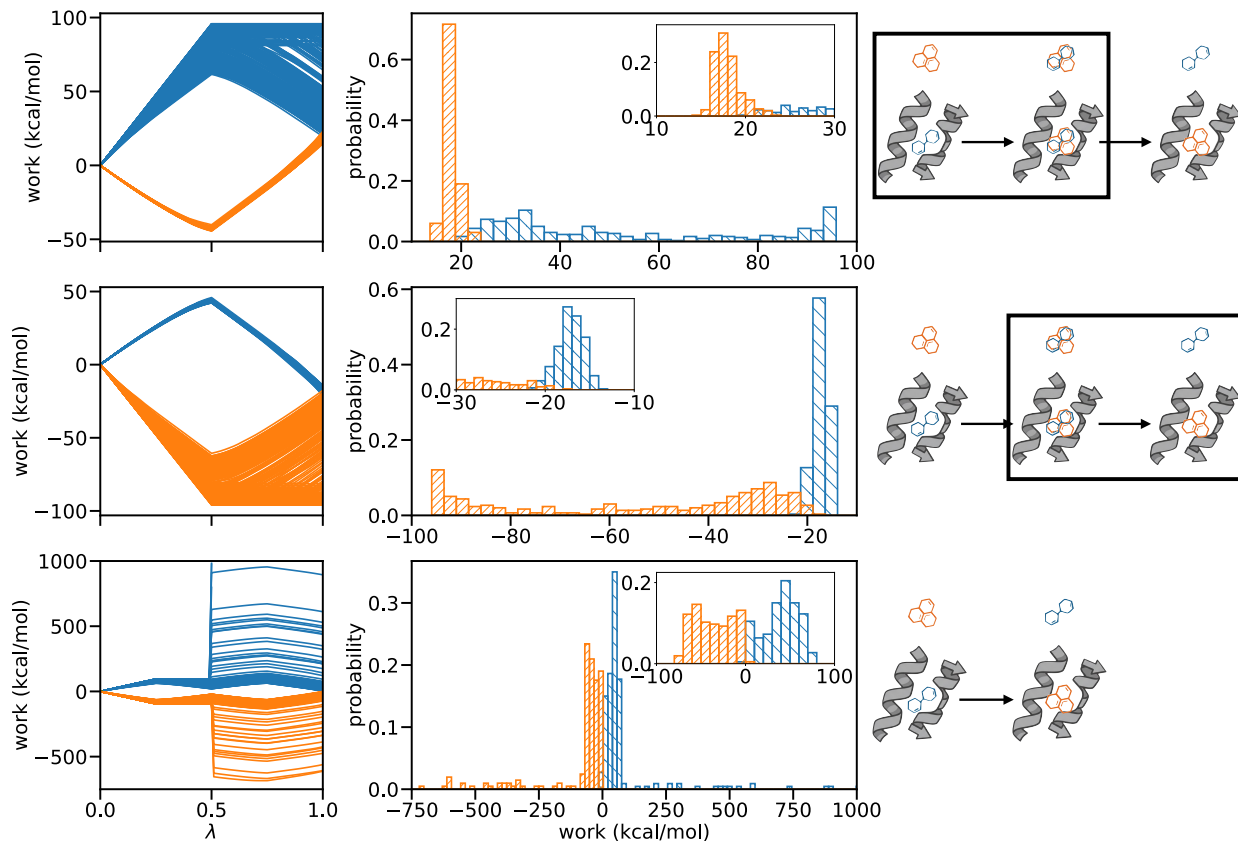


Figure 3: **Nonequilibrium chimeric switching exhibits significantly smaller work distribution variance than nonequilibrium nonchimeric switching.** Work trajectories (left), work distributions (center), and diagrammatic transforms (right) for a representative CDK2 ligand transformation (30 $\rightarrow$ 31). Forward and negative backward work are shown in blue and orange, respectively. **(A)** The first leg of the chimeric switch, transitioning from the initial physical state to the chimeric state. **(B)** The second leg of the chimeric switch, transitioning from the chimeric state to the final physical state. **(C)** The complete nonchimeric alchemical transformation, transitioning directly between the initial and final physical states without an intermediate.

Table 1: Statistical Comparison of Chimeric (M1) vs. Nonchimeric (M2) Methods. Δr and ΔRMSE represent the difference in performance (M1 - M2), where positive Δr and negative ΔRMSE indicate improvement by the Nonchimeric method. 95% confidence intervals are shown in brackets.

System	N	Δr [95% CI]	ΔRMSE [95% CI]	Wilcoxon p -value
CDK2	25	0.217 [0.068, 0.363]	-0.863 [-1.287, -0.422]	0.0074
TYK2	24	0.112 [-0.053, 0.278]	-0.189 [-0.410, 0.010]	0.3902
JNK1	30	0.051 [-0.061, 0.169]	-0.068 [-0.271, 0.134]	0.2286

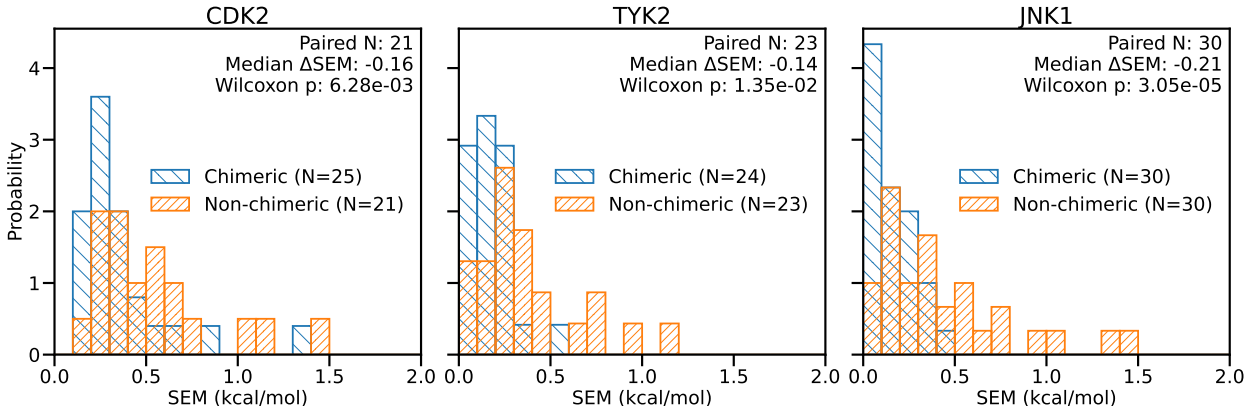


Figure 4: **Nonequilibrium switching with chimeric intermediates consistently recovers lower per-calculation error than nonchimeric switching.** Distributions of the standard error of the mean (SEM) for calculated $\Delta\Delta G_{\text{pred}}$. Chimeric and nonchimeric transforms are shown in blue and orange, respectively. **(Left)** CDK2 system. **(Middle)** TYK2 system. **(Right)** JNK1 system.

Chimeric states improve numerical stability with equivalent sampling time

Beyond predictive accuracy, the nonchimeric method exhibited signs of severe numerical instability, despite the doubling of the switching time to 200 ps. While the chimeric protocol maintained a failure rate of $\leq 2\%$ across all systems, the nonchimeric approach suffered from elevated rates of calculation failures (NaNs), specifically 16% for CDK2, 9% for JNK1, and 4% for TYK2. This suggests that the extended simulation time was an ineffective investment; the direct switching protocol struggles to locate high-probability phase space overlap in these complex systems, leading to numerical crashes rather than converged free energy estimates.

To investigate the mechanistic basis of this failure, we examined the nonequilibrium work trajectories for the transformation of Ligand 30 to Ligand 31 in the CDK2 system. This transformation serves as a representative case study for the instability observed in the nonchimeric protocol; for this specific edge, the nonchimeric method yielded a standard error of the mean (SEM) of nearly 3.0 kcal/mol (**Figure 2A**), a drastic increase compared to the tight precision of 0.16 kcal/mol obtained with the chimeric protocol (**Figure 1**). This degradation in precision was systemic: the nonchimeric results for CDK2 and JNK1 contained 3 and 10 edges, respectively, with SEM values exceeding 1.0 kcal/mol, whereas the chimeric method produced only 1 and 0 such edges, respectively.

Figure 3 compares the work trajectories and distributions for this representative transformation. The chimeric approach breaks the calculation into two constituent legs: transitioning from the initial physical state to the semi-interacting chimeric state (**Figure 3A**), and subsequently from the chimeric state to the final physical state (**Figure 3B**). In contrast, the nonchimeric method executes the full alchemical transformation directly without an intermediate (**Figure 3C**). It is evident that this nonchimeric switching method results in comparatively heavy-tailed work distributions. Furthermore, the work trajectories of the nonchimeric switches (**Figure 3C**, left) show large, nearly instantaneous variance accumulations around $\lambda = 0.5$. This variance accumulation and the resulting heavy-tailed distributions demonstrate a clear mechanistic reason for the degraded performance and numerical instability of the nonchimeric method.

Chimeric states improve free energy prediction precision

To systematically evaluate this degradation in precision across the datasets, we compared the distributions of the standard error of the mean (SEM) for calculated $\Delta\Delta G_{\text{pred}}$ (**Figure 4**). Free energy calculations for which forward and backward work distributions did not satisfy the overlap heuristic test were omitted from this analysis. For all calculations where overlap tests passed in both protocols, we performed a Wilcoxon signed-rank test. Across the CDK2,

TYK2, and JNK1 systems, the chimeric calculations showed statistically smaller SEMs with 95% confidence ($\Delta\text{SEM} = \text{SEM}_{\text{chim}} - \text{SEM}_{\text{nonchim}}$), confirming that the chimeric intermediate consistently recovers a lower per-calculation error than the nonchimeric approach.

NEX provides competitive scaffold hopping performance without additional parameter tuning

To investigate the performance of NEX on more difficult transformations, we benchmarked NEX on the BACE1 dataset curated by Bauman et al. for the equilibrium dual topology method Separated Topologies.²⁷ The BACE1 benchmark set contains 18 ligands from three series: an amide series, a spirocyclic series, and a biaryl series. This benchmark contains 45 transformations that map ligands within and across series. The resulting perturbation map **Figure S1** yields several challenging transformations within this benchmark set including scaffold hopping transformations, as illustrated in **Figure 5B**. We apply the protocol outlined in **Methods** section without modification and utilize the perturbation maps and input structures prepared by Bauman et al.²⁷

Figure 5A shows good correlation ($r = 0.71$ $CI_{95\%} : [0.53, 0.83]$) against experimental values. We note our performance is competitive with contemporary literature ($r = 0.91$ $CI_{95\%} : [0.85, 0.95]$, $\text{RMSE}_{\text{exp}} = 1.14$ kcal / mol). Importantly, to assess NEX’s ability to yield accurate predictions for scaffold-hopping transformations, all predictions were performed here with the same sampling as in our smaller lead optimization benchmarks. Further, we present our results here without fine-tuning, correction of individual predictions, and full automation of restraint selections. These results indicate that NEX supports accurate scaffold hopping transformations under a fully automated protocol, without reliance on manual intervention or post hoc correction.

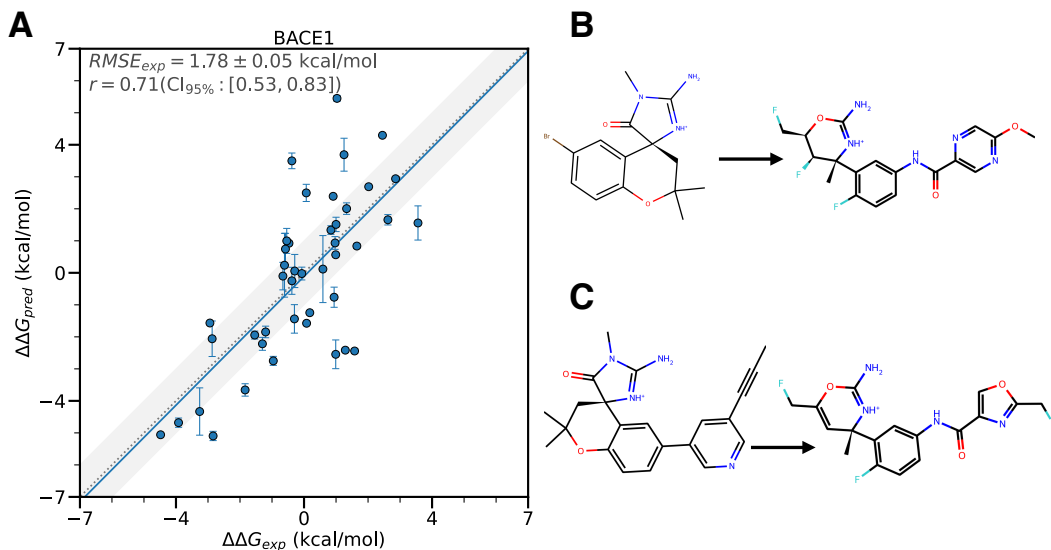


Figure 5: **Nonequilibrium switching offers competitive performance on BACE1 challenging benchmark set, including scaffold hopping transformations** (A) The predicted relative binding free energies ($\Delta\Delta G_{\text{pred}}$) on the y-axis are compared against the experimental relative binding free energies ($\Delta\Delta G_{\text{exp}}$) on the x-axis, both in kcal/mol for the 45 transformations connecting 18 ligands in the perturbation map. Each blue circle represents the average prediction from an individual relative binding free energy calculation, with error bars indicating the standard error of the mean (SEM) across three independent simulation runs. The dotted line shows the line of perfect agreement ($\Delta\Delta G_{\text{pred}} = \Delta\Delta G_{\text{exp}}$), and the solid blue line is the linear regression fit to the data. The shaded gray region highlights a ± 1.0 kcal/mol deviation from the ideal correlation. The inset in each plot provides the Root Mean Square Error with respect to the experiment (RMSE_{exp}) and the Pearson correlation coefficient (r), including its 95% confidence interval. All alchemical protocols were performed using the Alchemical Transfer Method. (B-C) Two representative edges from the benchmark illustrating challenging scaffold hopping transformations that are inaccessible via traditional single topology methods. **Figure S1** provides chemical structures for all 18 ligands and 45 edges connecting them in a perturbation map.

Discussion

We show that nonequilibrium chimeric switching provides an accurate, numerically stable framework for relative binding free energy (RBEF) calculations. Across multiple protein–ligand benchmarks, NEX achieves accuracy within the ± 1 kcal/mol regime while maintaining low estimator variance, high reproducibility, and numerical stability. A central result is that the inclusion of chimeric intermediate states is necessary for reliable nonequilibrium

estimation in a dual topology approach that unlocks chemical diversity. Direct end-to-end switching between physical states results in degraded accuracy, heavy-tailed work distributions, and elevated failure rates, even under increased switching time. In contrast, chimeric intermediates reshape the alchemical pathway to improve overlap between forward and reverse work distributions, thereby enhancing the statistical efficiency of estimators derived from nonequilibrium fluctuation relations. This level of reliability is a critical prerequisite for integration into agentic molecular design frameworks, where autonomous decision-making pipelines require consistently stable and failure-resistant predictions to operate at scale.

Beyond predictive performance, NEX offers several practical and methodological advantages. The use of equilibrium endpoint simulations to generate initial conditions for nonequilibrium switching naturally produces ensembles of configurations that can be retained and reused for downstream analysis. These trajectories enable detailed characterization of ligand binding modes, protein conformational responses, and solvent organization, including water network rearrangements,^{69,70} without additional simulation cost. Moreover, the framework is modular with respect to endpoint sampling, and emerging sampling approaches can be incorporated without modifying the nonequilibrium estimator.⁷¹

The ability to perform accurate, fully automated scaffold-hopping calculations is particularly valuable in drug discovery, where moving beyond local R-group optimization to explore new chemotypes is critical for improving potency and developability. Such transformations typically require expert tuning in RBFE workflows or require lengthy equilibrium sampling times,²⁷ limiting scalability and throughput. Automated, robust predictions enable high-throughput evaluation of diverse ligand series with fewer human in the loop interventions, improving reproducibility and accelerating decision-making in lead optimization and hit expansion.

In addition, double decoupling approaches used in numerous FEP methods rely on explicit alchemical interpolation of interaction potentials and require stable evaluation of partially interacting, nonphysical intermediate states.^{58,72} These requirements are difficult to satisfy

for machine-learned force fields, which are typically trained on fully interacting configurations and lack a natural decomposition into scalable energy components.⁷³ In contrast, the NEX framework operates directly on physical endpoint Hamiltonians, enabling straightforward integration with such models. This feature provides an additional avenue to optimize NEX performance in future work.

Conclusions

We introduce Nonequilibrium Chimeric Switching (NEX), a free energy framework that overcomes key limitations of alchemical methods by combining nonequilibrium statistical mechanics with chimeric intermediate states in a dual topology setting. By reshaping alchemical transformations to improve phase-space overlap, NEX enables accurate, low-variance relative binding free energy (RBFE) calculations across chemically diverse ligand pairs. We show that chimeric intermediates are beneficial for stable nonequilibrium estimation, yielding improved convergence and numerical robustness relative to direct switching approaches and extending the range of tractable RBFE transformations.

Although demonstrated here in RBFE applications focused on lead optimization and scaffold hopping, the NEX formalism is general and readily extensible. Future work will focus on absolute binding free energy (ABFE) calculations, early-stage hit identification,^{21,22,28,74} and more complex systems, such as peptides and biologics which can be structurally flexible,^{75–77} and charge changing transformations that should be amenable in this current NEX framework due to its adoption of ATM-like principles.^{24,25} The modular structure of NEX enables integration with enhanced sampling strategies and machine learning-based generative models,⁷¹ and its transfer-based formulation facilitates the incorporation of emerging force fields beyond the classical models used here for benchmarking.^{52,73}

By enabling efficient and numerically stable free energy estimation across chemically diverse transformations, NEX lays the groundwork for fully autonomous, agentic molecular

design systems, where stability is a prerequisite for scalable and reliable decision-making.

Supporting Information Available

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